PBAS Module Workflows

Budget Automation System

Step-by-step workflow guide for final year viva

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Project: PBAS-Frontend

1. Overall System Flow

1. User opens the system from the browser.
2. Splash screen appears first.
3. User clicks Login and enters email/password.
4. Backend verifies credentials and sends user details with token.
5. System checks user permissions.
6. Only allowed modules are shown in the sidebar.
7. User performs module actions according to role and permission.
8. Data is saved through REST APIs or local storage where frontend-only persistence is used.
9. Dashboard, reports, notifications, and audit logs show the result of system activity.

> **Login → Permission Check → Allowed Modules → User Action → API/Storage Save → Dashboard/Reports/Audit Update**

2. Login and Authentication Flow

1. User visits `/login`.
2. Splash screen shows system branding and login button.
3. User opens login form at `/login/form`.
4. User enters email and password.
5. Frontend sends login request to backend.
6. If email is not verified, system shows verification warning and resend option.
7. If login succeeds, token and user object are stored in Redux and localStorage.
8. User is redirected to Dashboard.
9. Protected routes validate token before showing private pages.
10. Logout clears auth state and sends user back to login.

> **Splash → Login Form → Credential Check → Token Save → Dashboard → Logout**

3. Dashboard Workflow

1. User opens Dashboard after login.
2. Frontend calls dashboard API.
3. System receives totals, charts, system health, and recent activities.
4. Cards show total budget, remaining amount, utilized amount, internal revenue, and pending approvals.
5. Budget chart shows remaining vs utilized budget.
6. Recent activities table shows request, payroll, budget, and notification statuses.
7. User can expand activity view to see more records.

> **Open Dashboard → Fetch Summary → Calculate Cards/Charts → Show Recent Activity**

4. User Management Workflow

1. Authorized user opens User Management.
2. System loads users from backend.
3. Admin/authorized user clicks Add User.
4. User details are entered: username, email, CNIC, father name, department, designation, role, and optional image.
5. Image is converted to Base64 if uploaded.
6. System sends create-user request to backend.
7. Backend creates user and emails credentials.
8. Existing user can be edited through edit action.
9. User account can be activated or deactivated with confirmation.

> **Load Users → Add/Edit User → Assign Role → Save → Activate/Deactivate**

5. Resource Request Management Workflow

1. Office Assistant or permitted user opens Resource Request Management.
2. User clicks New Request.
3. User selects request type: Government Grant, Internal Revenue, or Payroll.
4. User enters title, department, requester name, priority, amount, details, and attachment.
5. If request type is Government Grant, user can link a budget.
6. Request is created as draft.
7. Creator submits the request.
8. Admin Officer verifies submitted request.
9. Head of Institute approves or rejects verified request.
10. Audit trail can be viewed for request actions.

> **Create Draft → Submit → Admin Verification → HOI Approval/Rejection → Audit**

6. Budget Management Workflow

1. Authorized user opens Budget Management.
2. System loads all budget records.
3. User clicks Add Budget.
4. User enters budget name, start year, end year, amount, budget type, budget head, and budget category.
5. System checks duplicate category and overlapping year range.
6. If validation passes, budget is saved.
7. Budget can be edited if it is not archived.
8. Budget can be deleted if user has delete permission and year is not archived.
9. Search and year filter help find budget records.
10. Summary cards show total, remaining, utilized, and year-wise amounts.

> **Create Budget → Validate Year/Category → Save → Track Remaining/Utilized → Archive Lock**

7. Payroll Management Workflow

1. Accountant or authorized user opens Payroll Management.
2. Payroll data, staff master, cycles, and summary are loaded.
3. Staff Master stores employee details such as name, CNIC, post, BPS, basic pay, allowances, deductions, increment date, and increment rate.
4. Payroll sheet staff can be imported into Staff Master.
5. Annual payroll allocation is entered year-wise.
6. Payroll Estimates Management calculates estimated salary for staff.
7. Monthly payroll cycle is created by selecting month and year.
8. Payroll cycle is generated from active staff records.
9. HOI approves or rejects generated payroll.
10. After approval, payment status is recorded for staff items.
11. Payslip can be generated and printed.
12. At year end, payroll year can be archived and remaining balance is surrendered.

> **Staff Master → Estimates → Annual Allocation → Payroll Cycle → Generate → Approve → Payment → Payslip → Archive**

Payroll Estimate Calculation

Part	Formula
First 4 months	`monthly_basic_pay × 4`
Next 8 months	`monthly_basic_pay × 8`
Increment impact	`increment_rate × 6`
Total estimated salary	`first 4 months + next 8 months + increment impact`

8. Internal Revenue Workflow

1. Authorized user opens Internal Revenue.
2. User selects revenue month.
3. System loads saved revenue cells for selected period.
4. Spreadsheet shows BS Funds, Inter Funds, Registration Funds, Government Dues, Project Amount, and Grand Total columns.
5. User edits allowed cells only.
6. Formula columns are locked and calculated automatically.
7. BS total, Inter total, Grand Total, Total Expense, and Closing rows are calculated in frontend.
8. Changes are saved to backend through bulk upsert.
9. Rows are also stored locally for browser persistence.
10. User can export or print revenue reports.

> **Select Month → Edit Revenue Cells → Auto Totals → Save Cells → Export/Print**

9. Reports Workflow

1. User opens Reports module.
2. System loads report data from backend.
3. User selects period view: Weekly, Monthly, or Yearly.
4. User selects report type: All, Requests, Payroll, Budgets, or Notifications.
5. System creates generated report rows from backend status series.
6. Chart bars are calculated from each module count relative to maximum value.
7. Tables show recent requests, payroll records, notifications, and status breakdowns.
8. User can refresh reports.
9. User can export CSV report.

> **Fetch Reports → Select Period/Type → Generate Tables/Charts → Export CSV**

10. Audit Logs Workflow

1. Authorized user opens Audit Logs.
2. System loads audit records from backend.
3. User can filter by module, action, user, role, and date range.
4. User can search audit logs by text.
5. System formats raw audit actions into readable labels.
6. Internal revenue spreadsheet edits are summarized with changed cell count and preview.
7. Audit logs help examiner/user verify who performed which action and when.

> **Load Logs → Filter/Search → Review Actor/Action/Module/Time**

11. Notifications Workflow

1. User with notification permission sees notification icon in header.
2. System loads unread notification count.
3. User opens Notifications page.
4. Notifications show title, body, category, status, and optional link path.
5. User can mark one notification as read.
6. User can mark all notifications as read.
7. Unread count updates after read actions.

> **Load Notifications → Show Unread Count → Mark Read/All Read → Update Count**

12. Settings/Profile Workflow

1. User opens Settings from avatar/header.
2. System loads current profile from backend.
3. User edits name, email, father name, CNIC, department, and designation.
4. User can upload avatar image.

5. Avatar is cropped using cropper and canvas utility.
6. Profile update is saved to backend.
7. User can change password by entering current and new password.
8. After password change, backend may return updated token and user.

> **Load Profile → Edit Details/Avatar → Save Profile → Change Password**

13. Expense Management Workflow

Expense Management route currently exists as a placeholder module. The expected future workflow is:

1. Approved request becomes eligible for expense entry.
2. Accountant records actual expense amount and details.
3. System deducts expense from linked budget.
4. Expense appears in dashboard, reports, and audit logs.

> **Approved Request → Expense Entry → Budget Deduction → Report/Audit**

14. Role-wise Workflow Summary

User Role	Main Workflow
Head of Institute	Views dashboard/reports, approves or rejects requests, approves or rejects payroll cycles, checks audit logs and final system activity.
Admin Officer	Verifies submitted resource requests, may manage users and review operational records depending on permissions.
Accountant	Manages budgets, payroll staff, payroll estimates, payroll cycles, payments, payslips, annual allocation, and internal revenue.
Office Assistant	Creates and submits requests with amount, department, details, and attachments.

15. Final Viva Explanation Flow

1. Start by explaining that PBAS centralizes budget, payroll, request, revenue, reports, audit, and notification workflows.
2. Show login and explain token-based authentication.
3. Show role-based sidebar and explain that permissions decide visible modules.
4. Open Dashboard and explain live summary data.
5. Open Requests and explain draft to approval workflow.
6. Open Budget Management and explain budget creation, year/category validation, and remaining balance.
7. Open Payroll and explain Staff Master, Estimates, Payroll Cycle, Approval, Payment, Payslip, and Archive.
8. Open Internal Revenue and explain spreadsheet formulas and save workflow.
9. Open Reports and explain custom chart/table generation and CSV export.
10. Open Audit Logs and Notifications to explain traceability and alerts.